

SOFTWARE ENGINEER

Erin David Cullen

Business process automation and AI, built in Python and React

edcullen550@gmail.com
078 120 4717
Pretoria, South Africa
beardedseal77.github.io/CV
linkedin.com/in/de-cullen
github.com/BeardedSeal77

Profile

I build software around how a business actually runs: its processes, the people who carry them out, and the data that connects them. Python is my main backend language and React my frontend, and I'm also comfortable working in C# and Angular. Most of my work combines business process automation with AI.

That started in 2022 at Coastal Hire, where my parents own a franchise branch. I had worked the hire desk, the workshop and the logistics side, so I set out to automate the business's biggest pain points. That work became the Hire Operations Platform, now piloting in three branches. The tooling I built alongside it grew into DevTools, a workbench that gives developers and AI agents full visibility into a process-driven Python codebase.

Since January 2026 I've been a Software Engineer at BMW IT Hub South Africa, building an AI assistant for the department alongside C#/.NET systems and N8N automations. I'm finishing my BComputing dissertation this year. Before software I studied mechanical engineering, which still shapes how I think about systems.

Experience

Software Engineer
Jan 2026 – Present

BMW IT Hub South Africa

Sole developer of an AI assistant for the department, built from the ground up in Python for business analysts, agile masters and developers. It connects to Jira and Confluence and builds knowledge bases from them, so people can ask questions about a Confluence space and get useful answers. It also drafts sagas, epics and user stories that follow the team's Definition of Ready and Definition of Done.

On the product side, I built a category management system end to end, with a C# backend and an N8N automation, and it is now in production. I've also built a task system and an audit trail with structured logging in C# and .NET, following clean architecture with Entity Framework and MassTransit.

Full Stack Developer
2022 – Present

Independent Software Developer

Building software that automates how businesses run. My main work is the Hire Operations Platform, now piloting in three Coastal Hire branches. I gather requirements directly from the branches and their staff, and I own the full stack, from the PostgreSQL schema and Python backend to the React/Next.js frontend. Alongside it I built DevTools, a SQL-first workbench that lets developers and AI agents safely change and inspect a process-driven Python codebase.

In-House IT & Systems
2021 – 2023

Coastal Hire Centurion

In-house IT and operations for my family's Coastal Hire branch, part of a plant and tool hire franchise. Full-time on site from 2021: daily operations and management (including the high-pressure hire desk), technical and mechanical work in the workshop, and requirements analysis for the Kerridge ERP implementation. Moved to part-time in 2023 when studies at Belgium Campus began, with the systems work growing into the Hire Operations Platform. The company network runs on a Tailscale tailnet, and technical issues are still handled remotely on the odd occasion something goes wrong.

Junior Engineer
2021 (5 months)

Engineering Firm (under NDA)

Junior engineer at a mechanical engineering firm serving mining and heavy-industry clients (name withheld under NDA). Worked on part design and prototyping.

Skills

Backend & Data

Python · Flask · PostgreSQL · SQL Server · MongoDB · REST APIs · Microservices · Database Migrations · Business Process Modelling (BPM)

AI & Automation

Agent Design · MCP · Knowledge Bases · Jira & Confluence Integration · N8N · Ollama · OpenAI API · Whisper · Zero-Shot Classification

C# & .NET

C# · .NET · Entity Framework · MassTransit · Serilog · Clean Architecture

Frontend

React · Next.js · TypeScript · Tailwind CSS · Angular

Tools & Infrastructure

Docker · Docker Compose · Git & GitHub · Linux/WSL · Bash · Postman

Projects

Hire Operations Platform

Piloting in 3 branches

Business Process Automation • Hire Industry • Full-Stack

An operations platform for plant and tool hire, built to automate the biggest pain points of my parents' Coastal Hire franchise branch. It started from first-hand experience on the hire desk, in the workshop and in logistics, and covers the daily business processes around rentals, deliveries and customers.

Features: Diary Management · Logistics · Driver Scheduling · Task Management · Inventory Control · AI Automation · Rep & CRM · Debtors · Reporting · Data Aggregation & Validation · Database Auditing · Lock Control & Rollback

Impact: Piloting in three Coastal Hire branches with 10–15 users. Feedback from the branches drives each release, and wider rollout will follow once the pilot settles.

Python · Flask · PostgreSQL · React · Next.js · TypeScript · Docker

DevTools

In daily use

Developer Workbench • SQL-First • AI Agent Tooling

A self-hosted development workbench that gives developers and AI coding agents full visibility into a process-driven Python codebase. It treats the database schema as versioned documents in a git-backed journal: each change is a single operation, applied to the development database straight away and compiled into a reversible migration. It reads each service's declared actions, events and processes and draws the architecture and business-process flows as live diagrams. Its planning board links every story and bug to the feature documentation, the commits and the AI cost spent on it, so each requirement can be traced from spec to code to test.

Features: Schema as versioned documents · Reversible migrations, data preserved · Live architecture & BPM diagrams · Scenario tests with fixtures · Data-quality & health checks · MCP server for AI agents · Portable AI-agent setup · Planning board: sagas to bugs · Spec-to-code traceability

Impact: Developers and AI agents describe the change they want and DevTools writes the SQL, so nobody writes migrations by hand and an AI never touches the database directly.

Python · Flask · Next.js · MongoDB · Docker · MCP

Local AI Chat Assistant

Local AI • Privacy-Focused • Tool Use

A chatbot built on Python and Ollama that runs entirely on local hardware. It can write code and browse the web, and keeps a long-term memory in PostgreSQL.

Features: Runs fully locally for privacy · Writes and explains code · Browses the web for answers · PostgreSQL-based memory

Ollama · Python · PostgreSQL · NLP

Nimbus: NLP Drone Control

Natural Language Processing • Computer Vision • Mixed Reality

An intelligent drone navigation system enabling natural language control in mixed reality environments. Voice commands like "Go to the chair" are carried out autonomously using computer vision and SLAM technology.

Features: Modular Python architecture with containerized services · Docker containers for NLP, vision and control modules · Real-time multi-threaded processing pipeline · Microservice communication via REST APIs

OpenCV · NumPy · NLTK/spaCy · PyTorch · Docker · Flask/FastAPI

github.com/BeardedSeal77/Nimbus

Education

Bachelor of Computing (Data Science)

Belgium Campus

2023 – Present · To complete end of 2026

Focusing on data science and database systems, developing expertise in modern software development and data analysis techniques.

Mechanical Engineering

University of Pretoria

2018 – 2021

Award: Most Innovative Project (EBIT Faculty)

Studied mechanical engineering for four years but did not complete the degree, choosing to move into software instead. The engineering mindset still shapes how I approach systems design and problem-solving.

National Senior Certificate

St. Albans College (IEB)

Certificates

2024	Mensa Member, Pretoria Chapter
2021	Zeiss Geometric Dimensioning & Tolerancing
2019	Most Innovative Project (EBIT Faculty, University of Pretoria)
2016	Dale Carnegie Generation.Next

Qualities

Accurate & Precise · Detail-Oriented · Analytically Minded · Self-Motivated · Always Learning

Hobbies

Carpentry & Woodworking · Hiking & Nature · Photography

References

Available on request